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Short-term impact assessment of ocean liming: A copepod exposure test

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ABSTRACT

Ocean liming (OL) is a potential carbon dioxide removal (CDR) method that aims to increase the ocean's capacity to absorb atmospheric CO₂ by adding hydrated lime to the surface ocean. Modeling studies indicate that OL may cause temporary pH spikes lasting several minutes, depending on the lime sparging rate. Little is known about the short-term effects of these spikes on marine organisms. Aim of the present study is to investigate these effects on the copepod *Acartia tonsa*. Copepods were exposed to different pH conditions (9, 10, 11, 12) by dosing different hydrated lime solutions. Copepod mortality, movements, and behavior were recorded. At pH 9 for short exposure times (<6 h), no negative effects were observed indicating a potential tolerable threshold for OL applications. At longer exposure times (>6 h) and pH higher than 9, negative effects (mortality and sublethal effects) were found significantly higher than in the control.

The most recent IPCC assessment has shown that keeping global warming well below 2 °C requires drastic and rapid emission reductions, as well as the deployment of carbon dioxide removal (CDR) technologies (IPCC, 2023). In order to be effective, a portfolio of CDR methods is called to extract from the atmosphere several hundred gigatons of CO₂ and store them safely and for long in other carbon reservoirs (i.e. impermeable caprock over porous sedimentary rocks) (Kampman et al., 2016; Rogelj et al., 2018). One potential CDR method is Ocean Alkalinity Enhancement (OAE). The idea behind OAE is to increase the ocean's chemical storage capacity for atmospheric CO₂ by adding proton-neutralizing substances to the surface ocean (Kheshgi, 1995). Enhanced alkalinity causes a shift in the inorganic carbon speciation in seawater, from carbon dioxide (CO₂) to bicarbonate (HCO₃⁻) and carbonate (CO₃²⁻), thereby making new space for additional atmospheric CO₂ to be absorbed (Hartmann et al., 2013). In addition, OAE may counteract ocean acidification, which is considered a threat to marine ecosystems (Doney et al., 2020). OAE can be achieved through various approaches (Renforth and Henderson, 2017). Most of these are either directly or indirectly linked to the chemical weathering of minerals, such as limestone-derived products.

Caserini et al. (2022) showed that slaked lime (e.g., calcium hydroxide), due to its lower content of potentially toxic elements, may be the most appropriate option for OAE also considering its widespread availability and the fewer environmental concerns compared to other minerals, such as limestone, olivine, magnesite, and brucite. However, besides metal toxicity, impacts on marine ecosystem are largely unknown, especially regarding alkalization process (Kohler et al., 2013; Hauck et al., 2016; Taylor et al., 2016; Feng et al., 2017; Ferderer et al., 2022). Caserini et al. (2021a) used a simplified modeling approach to simulate the fluid dynamics of slaked lime sparging into the ocean from a ship. The aim of this study was to define the relationship between slaked lime sparging rate, and the corresponding pH increase in seawater. Based on their simulations, the effect on seawater pH, was expected as a pH increase of approximately 1–1.5 units lasting only a few minutes. Understanding whether these short-lasting pH spikes could substantially affect marine organisms is challenging based on current literature since it principally addresses the relationship between climate change and ocean acidification. Only recently, research papers discussing ocean alkalization have emerged. Among the limited number of authors who have explored the effects of high pH on natural marine

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communities, Pedersen and Hansen (2003) stand out. Their study demonstrated that prolonged elevated pH levels (>9) in natural settings can profoundly influence the entire plankton community, primarily by reducing species diversity and favoring algal blooms due to a decline in grazing pressure. Gim et al. (2018) conducted a literature review of seawater acidification studies, indicating a significant dose-response relationship between pH and its ecotoxicological effects on 19 marine organisms. Among these, the copepod *Tigriopus japonicus*, Mori, 1938 showed the highest mortality rate and was the most sensitive endpoint at the highest pH exposure. It should be also noted that the majority of toxicity tests involve predetermined exposure times, typically ranging from 24 to 48 h, in order to evaluate the response to varying levels of the stressor being studied. However, concerning the potential effects of OL-induced pH peaks, the exposure times of interest are shorter than 24 h, and this information is missing in the available literature.

Therefore, the aim of this study is to analyze the short-term effects (<24 h) on zooplankton of the pH peaks which may potentially result from slaked lime sparging into the ocean.

Due to the lack of a standardized procedure for the specific objective of the test (i.e., short-term exposure, below 24 h, to pH levels higher than the typical conditions), the acute copepod tests were designed broadly following the guidelines provided by ISO 14669 (International Organization for Standardization, 1999). As test species *Acartia tonsa* Dana, 1849 was selected being a common and widely distributed species, found in various marine habitats, including the Mediterranean Sea. Due to its sensitivity to changes in environmental conditions, *A. tonsa* is among the species recommended by the International Organization for Standardization (ISO) for acute assessment of marine contaminants (ISO/TC147/SC5) and toxic substances, it is therefore widely accepted in regulatory frameworks for ecotoxicological testing. In addition, *A. tonsa* is the food source for many marine organisms and it is relatively easy to culture in the laboratory in large quantities, having a rapid life cycle. Adult specimens of *A. tonsa*, culture water and algal mix feeding were purchased from Guernsey Sea Farms Ltd. (Port Vale, Guernsey, UK). Copepods were cultivated at salinity 35, and after their arrival, they were transferred immediately to our climatic culture room and kept in 5-L glass beakers, with fresh culture water and under gentle continuous aeration. The organisms were kept at room temperature ($20 \pm 1^\circ\text{C}$), under 16/8 h light/dark cycles, and fed with a mix of algae (*Tetraselmis suecica*, *Pavlova lutheri* and *Tisochrysis lutea*) for 24 h only before the experiments. Due to the technology's global reach, we decided to test a wide range of pH levels. So following Pedersen and Hansen (2003), who tested natural pH ranges, we selected pH 9 as the most probable value after OL sparging, pH 10 as a possible local natural maximum (e.g., coastal areas with a high Net Primary Production). Furthermore, assuming an increase of 1–1.5 units beyond the typical natural extreme of pH 10, we conducted tests at pH levels of 11 and 12. Adults and juvenile individuals (i.e. copepodites), were exposed to selected pH conditions to assess the effects on their mortality, motility or any evidence of behavioral changes. To obtain the pH conditions to be tested, ultrapure hydrated lime powder ($\text{Ca}(\text{OH})_2$ product specifications are summarized in Table S1) was dosed in 1 L of culture water (Nalgene bottle). Bottles were stored in the dark at culture room temperature.

Tests were performed on adults and copepodites randomly collected from the culture, one by one, using a micropipette (p-5000) with a modified tip with a widened opening to prevent harm to the organisms (Table S2).

Copepod stage and sex, whenever feasible, were determined under the stereoscope. A total of 180 organisms for each pH level condition experiment were used. However, due to some issues encountered during the shipping phase, the number of organisms employed in the pH 10 experiment was reduced from 180 to 120.

For each pH value, three replicas of control and three replicas of the test have been done. Just before the test, $\text{Ca}(\text{OH})_2$ solution (test) or culture water (control) was distributed in 12-well plate dishes, filling 10 wells with 5 ml and placing a single individual.

Four levels of response variables (status) were identified:

- (1) *Dead*. Individuals still lying at the bottom of the plate and not moving after repeated gentle stimuli (i.e., no swimming and no appendage movement for 10s as described in ISO 14669)
- (2) *Motionless*. Individuals showing no movements and moving or swimming only after being gently prodded with a tip for 10s
- (3) *Lethargic*. The individuals swimming slowly and lethargic
- (4) *Alive*. The individuals actively swimming

The number of individuals corresponding to each level was counted through a stereoscope at 5 min, 3 h, 6 h, 24 h and 48 h after the beginning of the experiment. The test duration was extended to 48 h to account for lethal effects which can be delayed (Beketov and Liess, 2008).

The statistical association between the four status and pH conditions was assessed through a chi-square testing procedure and Spearman rank correlation coefficient. Moreover, to identify the main factors of variability, the four status categorical variable was recoded into a binary one (e.g., *alive* vs *impaired*) and binary logistic regression analysis was applied with reference target category of the *impaired*. Furthermore, alive individuals were balanced with the number of impaired individuals through a weighting procedure. A stepwise method based on the Wald statistic was used (Hosmer and Lemeshow, 2000) in order to select the best set of predictors. In the forward stepwise procedure, each predictor is tested for entry into the model one by one, based on the significance level of the Wald statistic. The procedure stops when no more variables meet the entry criteria or when the final model is the same as the previous one. The Control/Tested factor enters at the first step, as the most relevant predictor, followed by pH, exposure duration and life stage. In logistic regression, the relative importance of each predictor can be assessed through the Exp(B) term which is also known as the odds ratio. Odds ratio indicates the change in odds resulting from a unit change in the predictor. Finally, in order to assess the effect sizes of the observed patterns of toxicity a Generalized Linear Model (hereinafter called GLM, see Rutherford, 2011) was used:

$$y_{ijk} = \mu + \text{xtime_hrs} + \alpha_i + \beta_j + k + \alpha\beta_{ij} + \beta\gamma_{jk} + \alpha\gamma_{ik} + \alpha\beta\gamma_{ijk} + \varepsilon_{ijk}$$

where: y_{ijk} : percent of individuals showing adverse effects (% impaired), μ : true overall mean, xtime_hrs : covariate to account for the exposure duration, α : incremental effect of Test vs Control, β : incremental effect of pH (9, 10, 11), γ : incremental effect of life stage (e.g., copepodites (C) vs adults (A)), $\alpha\beta_{ij}$: Control/Test * pH interaction term, $\beta\gamma_{jk}$: pH * life stage interaction term, $\alpha\gamma_{ik}$: Control/test * life stage interaction term, $\alpha\beta\gamma_{ijk}$: control/Test * pH * life stage interaction term, ε_{ijk} : residual error.

The impact of the exposure to $\text{Ca}(\text{OH})_2$ enriched solutions (pH = 9, 10, 11, 12) on *A. tonsa* mortality/sublethal effects were tested at different times: 5 min, 3, 6, 24 and 48 h (Table 2). At pH 12 all the tested individuals died instantly, therefore the trial at pH 12 was not included in the following analysis. Effects were observed on copepod mobility mostly at pH condition >9 (Fig. 1).

Chi-square test revealed a highly significant association between pH conditions and copepod status (chi-square: 603.2, df: 6; $P < 0.001$). Moreover, being both pH and status e.g., from alive (0) to dead (3), ordinal variables, the Spearman rank correlation coefficient was also evaluated outlining a significant positive correlation between the two (Spearman's rho: 0.376, $P < 0.001$).

In accordance with ISO 14669, any additional observed effect on copepod mobility (such as lethargy or immobility) was deemed adverse. This is because impaired mobility would significantly increase their susceptibility to predators and hinder their ability to forage successfully.

No adverse effects were observed in the control plates within the first 5 min, as well as at 3 and 6 h. However, for longer times (e.g., 24 and 48 h), some (10–20 %) mortality was observed for both copepodite and adult life stages (see Fig. 2).

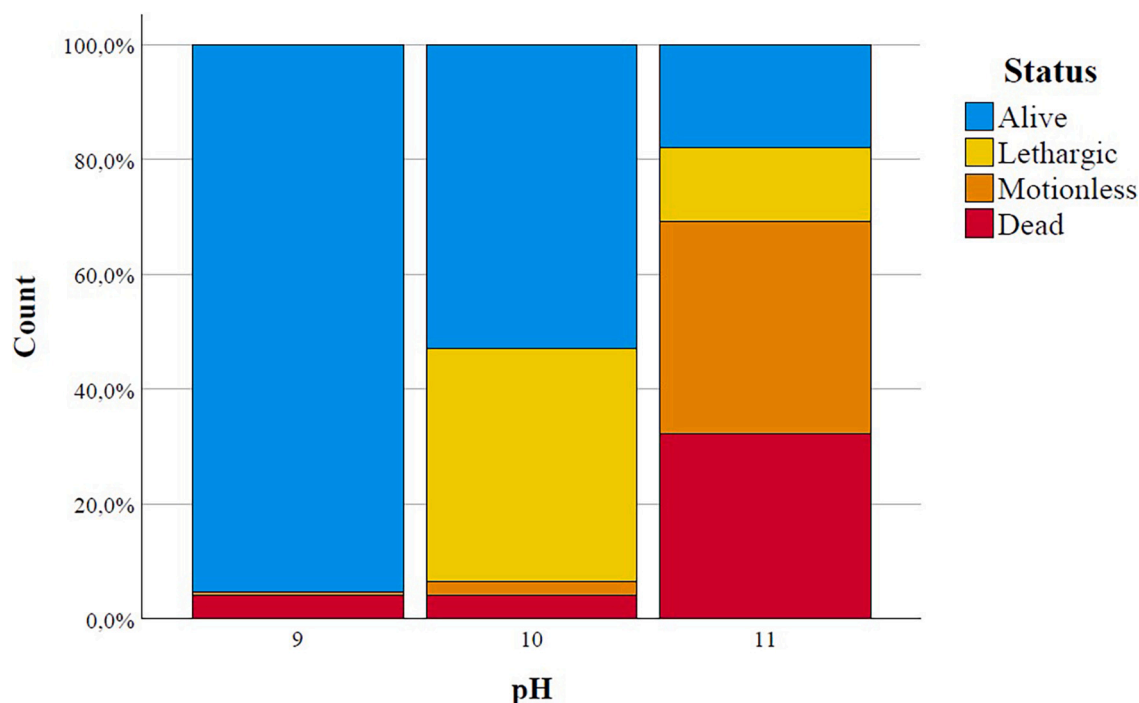


Fig. 1. Statistical association between pH conditions and copepod status (cumulative effects over the whole exposure duration).

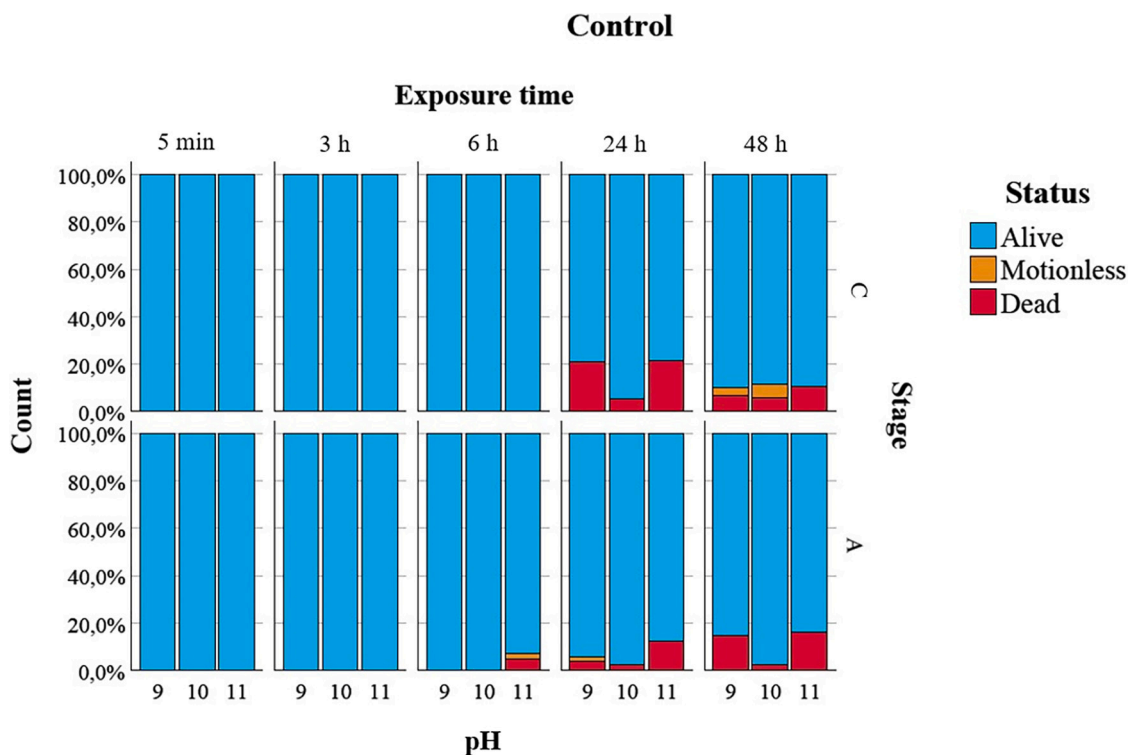


Fig. 2. Copepods' status observed in the control plates.

In the treatment plates (Fig. 3) mortality and sublethal effects clearly increased.

At pH 9 and exposure durations of up to 6 h, no or minimal detrimental impacts were noted, whether in relation to copepodite (C) or adults (A). However, adverse effects were evident at exposure periods shorter than those observed in control plates, particularly concerning pH levels of 10 and 11.

Life stage seems to be an important factor of influence, being copepodite stages more sensitive to pH increase.

In order to better investigate the exposure-effects relationship, a binary response variable, was created categorizing individuals as either alive or impaired, and subjected to binary logistic regression analysis. The resulting regression models are presented in Table 1.

As shown in Table 1, the Control/Test factor emerges as the most

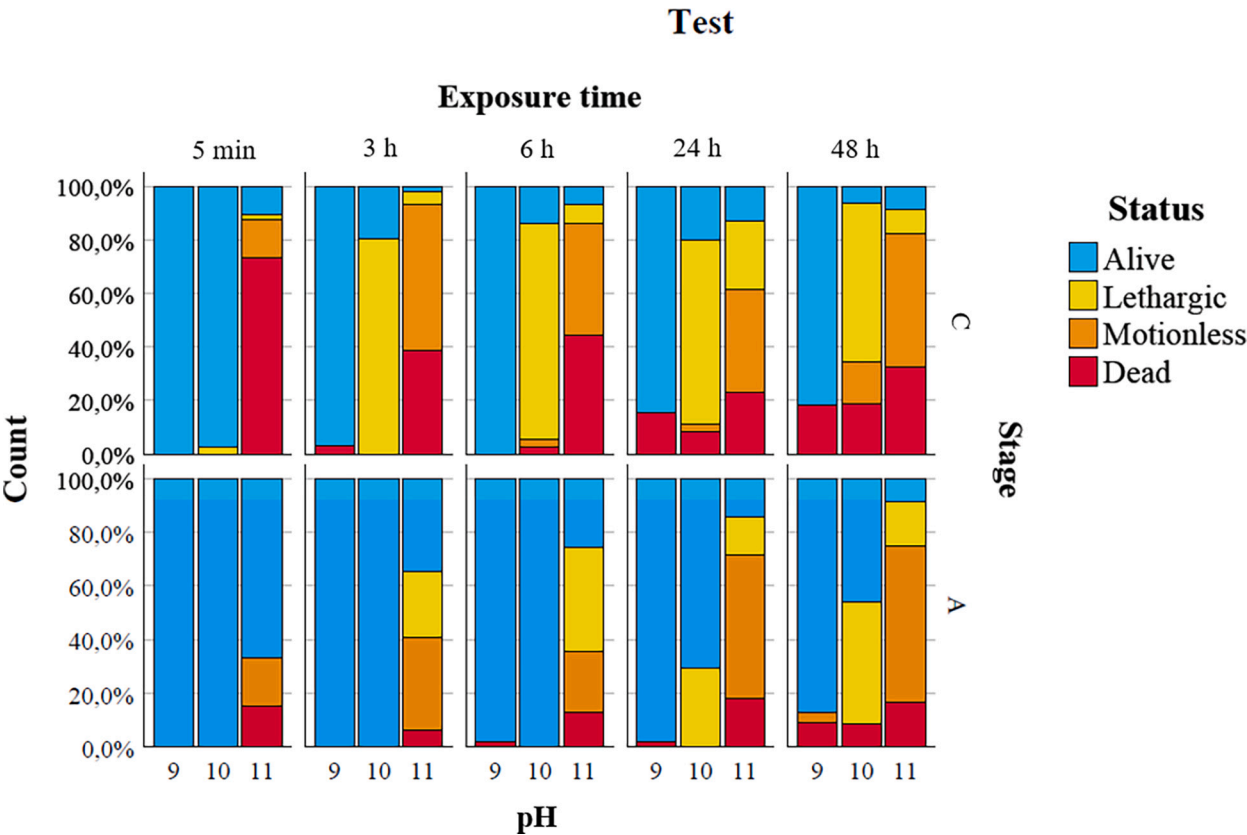


Fig. 3. Copepods' response to pH peak exposure in tested plates.

Table 1
Results of the stepwise logistic regression analysis. Regression coefficients (B) and corresponding standard errors (S.E.), Wald statistics and corresponding *P*-values are shown. Odds ratios (Exp(B)) and corresponding 95 % confidence interval are also shown.

Variables in the equation		B	S.E.	Wald	df	P-values	Exp(B)	95 % C.I. for EXP(B)	
								Lower	Upper
Step 1 ^a	Control/test	2.52	0.17	214.37	1	<0.001	12.45	8.88	17.45
	Constant	−1.72	0.15	132.31	1	<0.001	0.18		
	pH	1.30	0.11	143.19	1	<0.001	3.66	2.96	4.53
Step 2 ^b	Control/test	2.73	0.19	196.34	1	<0.001	15.32	10.46	22.44
	Constant	−15.16	1.16	169.28	1	<0.001	0.00		
	Exposure time	0.03	0.01	40.40	1	<0.001	1.03	1.02	1.04
Step 3 ^c	pH	1.41	0.11	150.83	1	<0.001	4.11	3.28	5.15
	Control/test	2.93	0.21	197.96	1	<0.001	18.67	12.42	28.08
	Constant	−17.04	1.27	180.16	1	<0.001	0.00		
Step 4 ^d	Exposure time	0.03	0.01	43.95	1	<0.001	1.03	1.02	1.04
	pH	1.36	0.12	134.90	1	<0.001	3.88	3.09	4.88
	Control/test	2.86	0.21	182.29	1	<0.001	17.44	11.52	26.42
	Life stage	−1.00	0.18	31.82	1	<0.001	0.37	0.26	0.52
	Constant	−16.02	1.29	154.26	1	<0.001	0.00		

^a Variable(s) entered on step 1: Control/Test.
^b Variable(s) entered on step 2: pH.
^c Variable(s) entered on step 3: Exposure time.
^d Variable(s) entered on step 4: Life Stage.

significant predictor, entering step 1, followed by pH, exposure duration, and life stage. Even in the step 4 model, the Control/Test factor remains the strongest direct predictor, with pH and exposure duration as subsequent predictors. Notably, life stage exhibits a significant inverse correlation, indicating a reduced risk of impairment associated with adult life stages. Sex, on the other hand, did not prove to be a significant factor of influence and was therefore excluded from subsequent analyses. Overall, the four logistic models exhibit remarkably high

accuracy, ranging from 74.7 % in the step 1 model to 85.4 % in the step 4 model (refer to Table 2).
Table 3 presents the effect sizes and significance levels estimated through the Generalized Linear Model, which correlates the same factors identified in the binary logistic model with the percentage of impaired copepod individuals.
Consistent with the findings of the logistic regression analysis, the Control/Test factors exhibit the largest effect size, followed by pH,

Table 2

Confusion matrix showing the classification performances of the logistic models.

Classification table ^a					
Observed			Predicted		
			Status 01		Percentage correct
			0	1	
Step 1	Status 01	0	295	195	60.2
		1	53	437	89.2
	Overall percentage				74.7
Step 2	Status 01	0	410	80	83.6
		1	72	418	85.3
	Overall percentage				84.5
Step 3	Status 01	0	367	123	75.0
		1	50	440	89.8
	Overall percentage				82.4
Step 4	Status 01	0	410	80	83.7
		1	63	427	87.1
	Overall percentage				85.4

^a The cut value is 0.500.

Table 3

Effect sizes determined through the GLM analysis: deviance (SS) and variance (MS) components are shown as the corresponding F statistics and P-values (P).

Tests of between-subjects effects					
Dependent Variable: % impaired Individuals					
Source	SS	df	MS	F	P
Corrected model	146,817.9 ^a	12	12,234.8	40.035	0.000
Intercept	26,386.8	1	26,386.8	86.344	0.000
Exposure time	6701.3	1	6701.2	21.928	0.000
Control/test	56,332.0	1	56,332.0	184.332	0.000
Life stage	6095.1	1	6095.1	19.945	0.000
pH	32,471.7	2	16,235.8	53.128	0.000
Life stage * Control/test	5106.9	1	5106.9	16.711	0.000
pH * Control/test	28,517.1	2	14,258.5	46.657	0.000
Stage * pH	3018.1	2	1509.0	4.938	0.009
Stage * pH * Control/test	2806.8	2	1403.4	4.592	0.012
Error	36,366.5	119	305.6		
Total	273,114.0	132			
Corrected total	183,184.4	131			

^a R squared = 0.801 (adjusted R squared = 0.781).

Exposure time, and Life Stage. The interaction effects are illustrated in Figs. 4 and 5, where it can be observed that in the Control plates, no significant effect was detected for both pH conditions and life stages (Fig. 4). However, in the tested plates, significant effect sizes were observed for both Life Stages and pH conditions (Fig. 5). Furthermore, at a pH of 9 and exposure time below 6 h, no adverse effects were observed in the tested plates, indicating no significant difference compared to the Control plates (Fig. 3).

The limited knowledge of the Ocean Alkalinity Enhancement impacts on marine ecosystems may affect the design of sparging methods, altering the capability to mitigate potential adverse effects (e.g. the temporary pH peaks, Caserini et al., 2021b; Ferderer et al., 2022). It is in fact well known that pH changes may significantly impact planktonic organisms (Pedersen and Hansen, 2003; Ferderer et al., 2022). However, specifically designed experiments are required to address this issue. Literature can provide limited information since the majority of existing studies focus on acidification rather than alkalization impacts and have tested exposure times higher than the few-minutes-interval predicted by models (such as Caserini et al., 2021a). In aquatic systems, zooplankton organisms, and in particular copepods, dominate the pelagic food web that is very sensitive to induced variations, both of natural and anthropogenic origin (Camatti et al., 2023); so, any important modification in terms of copepods' abundance and dimensional structure may alter the food web connectivity, the efficiency of trophic transfer, and its own contribution in terms of ecosystem services (sensu Costanza et al., 1997). The experiments we have conducted, focusing on the copepod species *A. tonsa*, shed light on the potential impact of pH variation on copepod mobility and survivorship. In the experiments the most pronounced effects both on mortality and sublethal effects were observed at pH conditions above 9.

This is consistent with the results of the Pedersen and Hansen, 2003 who observed at the laboratory scale some toxicity effects on copepods community at pH conditions of about 9.

Life stage was found to be a critical influencing factor, as the copepodite stage appeared to be more sensitive to pH increases than the adult stage. It is worth noting that at pH 9 and exposure times shorter than 6 h, no adverse effect was observed when compared to control conditions, suggesting a potential threshold where copepods can tolerate short-term pH changes. Copepods in the ocean exhibit notable seasonality in their life cycles, with population dynamics and reproductive cycles closely

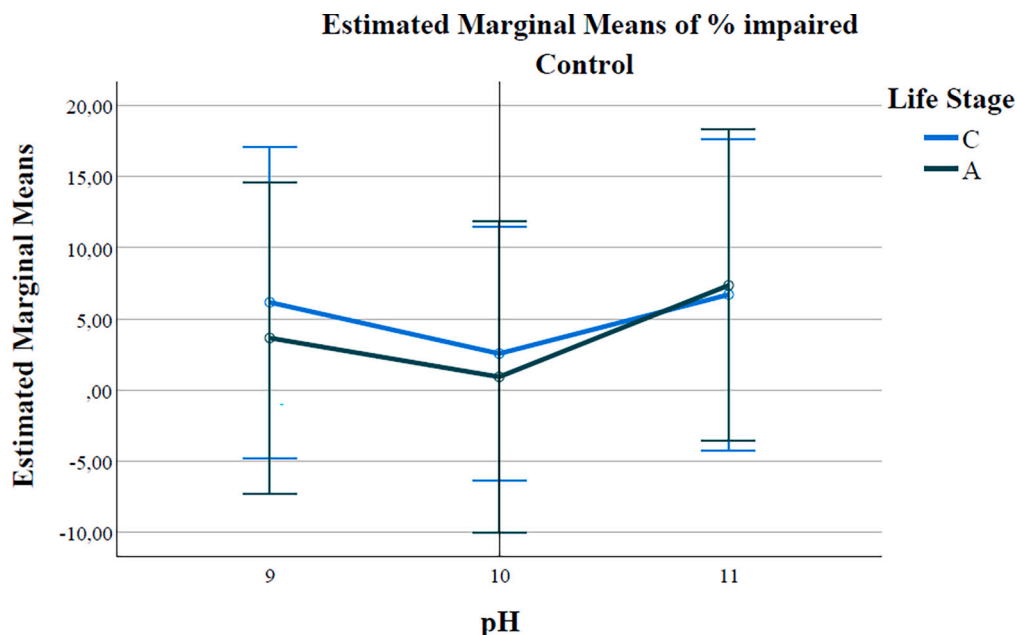


Fig. 4. Interaction terms effects: in the Control plates no significant effect could be assessed for both pH conditions and life stages.

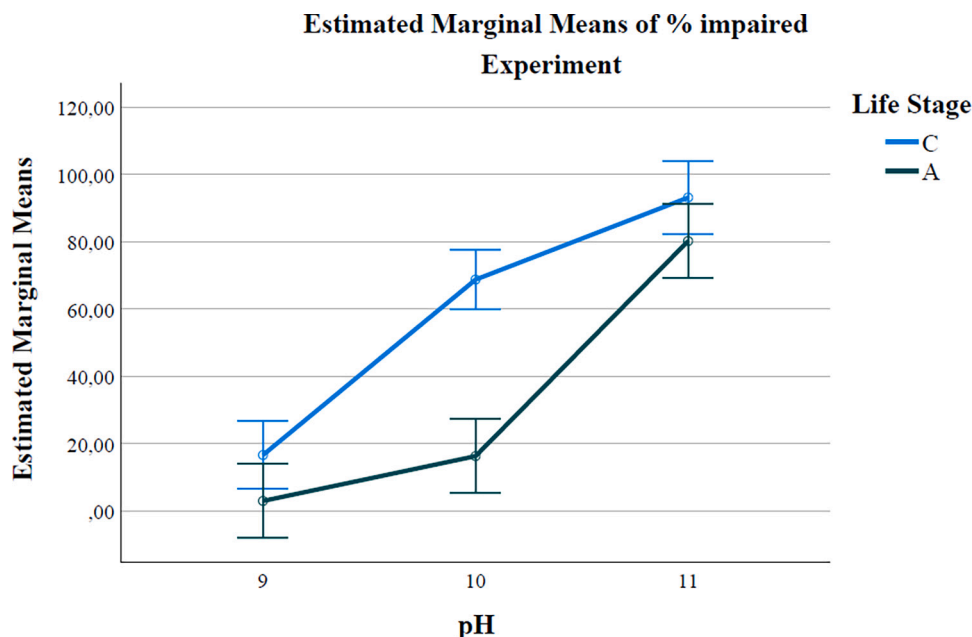


Fig. 5. Interaction terms effects: in the Tested plates significant effect sizes were observed for both life stages and pH conditions.

tied to seasonal variations (Kosobokova and Hirche, 2000; Beaugrand, 2004). Environmental factors such as temperature, food availability, and changes in the surrounding conditions during different seasons play key roles in shaping copepod life cycles (Hirst and Bunker, 2003). The timing of critical life cycle events, such as egg production, hatching, and the development of various life stages, can vary in response to seasonal changes in temperature and nutrient availability. These fluctuations in copepod populations may have significant implications for the marine food web (Costanza et al., 1997) and biogeochemical cycles (Dam et al., 1995; Buitenhuis et al., 2006), and can trigger cascading effects on other organisms within the ecosystem (Sato and Hirche, 2007). Given the pronounced seasonality and life cycle dynamics observed in copepods, it is crucial to carefully consider these aspects and their implications when designing an Ocean Alkalinization Enhancement experiment. Particularly a better understanding would be needed on the environmental factors (such as temperature, nutrient availability, and other seasonal fluctuations) which may influence the effects of alkalinization on copepod populations and their associated ecosystems.

In conclusion, the experiments conducted on the copepod *Acartia tonsa* have yielded valuable insights into the short-term effects of pH conditions on copepod mobility and survivorship.

However, it is essential to note that these conclusions are based on short-term experimental data in controlled conditions and cannot be extrapolated to predict longer-term effects of pH spikes. Nevertheless, they provide crucial insights for the design of Ocean Liming processes. Considering the broader implications of affecting a vital component of the marine food web, it is imperative to consider the natural seasonal variability of the zooplankton component when developing and implementing OL initiatives. By doing so, potential risks can be mitigated by maintaining pH variability below the precautionary threshold of 9 and adopting a cautious approach that accounts for natural fluctuations. Future studies are certainly needed to improve our understanding of OL potential adverse effects on zooplanktonic species. Given their vast taxonomic diversity and key role in the marine food web, conducting experiments at a mesocosm-scale could significantly expand our understanding to encompass other biological compartments.

CRedit authorship contribution statement

Elisa Camatti: Conceptualization, Data curation, Investigation,

Methodology, Resources, Supervision, Validation, Writing – review & editing. **Silvia Valsecchi:** Data curation, Investigation, Methodology, Resources, Validation, Writing – review & editing. **Stefano Caserini:** Funding acquisition, Project administration, Validation, Writing – review & editing. **Eleonora Barbaccia:** Data curation, Formal analysis, Visualization, Writing – original draft, Writing – review & editing. **Chiara Santinelli:** Investigation, Methodology, Resources, Supervision, Writing – review & editing. **Daniela Basso:** Conceptualization, Funding acquisition, Methodology, Project administration, Supervision, Writing – review & editing. **Arianna Azzellino:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Methodology, Project administration, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.marpolbul.2023.115833>.

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